

Technical Document #701: Cloud Computing - Comprehensive Overview

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Infrastructure as a Service (IaaS) provides virtualized computing resources over the internet. AWS EC2 and Azure VMs are popular IaaS offerings.

Kubernetes orchestrates containerized applications across clusters of machines. It handles deployment, scaling, and management of containerized workloads.

Containerization packages applications with their dependencies. Docker creates lightweight, portable containers that run consistently across environments.

Platform as a Service (PaaS) provides a platform for developing and deploying applications. Heroku and Google App Engine are PaaS examples.

Serverless computing allows developers to run code without managing servers. AWS Lambda and Azure Functions automatically scale based on demand.

Content delivery networks (CDNs) distribute content globally to reduce latency. CloudFront and Cloudflare are popular CDN providers.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Key Takeaways:

- Auto-scaling automatically adjusts compute resources based on demand.

- Infrastructure as a Service (IaaS) provides virtualized computing resources over the internet.
- Content delivery networks (CDNs) distribute content globally to reduce latency.